MCQ solutions

1. **B.** The sign $f'' > 0$ means that f' increases, hence the convexity. It does not give the sign of f' nor that of f.
2. **C.** The derivative of $e^{u(x)}$ is $u'(x)e^{u(x)}$. Here $u(x) = 1 + x^2$, therefore $u'(x) = 2x$. Forgetting this factor is a error in composition.
3. **B.** A differentiable convex function lies above each of its tangents on the interval of convexity. These tangents need not be horizontal.
4. **B.** The positive sign of f' then negative gives an increase then a decrease around a : maximum local, and not a minimum.
5. **C.** $|x| \to 0$ when $x \to 0$, hence continuity. The left and right slopes are -1 and 1, therefore not of differentiability at 0.
6. **A.** The IVT uses continuity on the whole interval. No zero derivative is required; an interval reduced form gives no change of sign.

7. **B.** Continuity and a change of sign guarantee existence. A function may have several roots: uniqueness requires an additional argument.
8. **B.** A strictly monotonic function is injective: a given value has at most one preimage. Being bounded or even does not guarantee this property.
9. **B.** As long as no exact root is found, we keep the sub-interval whose endpoints have images of opposite signs; its length is halved.
10. **B.** Over an interval where $f'' < 0$, the derivative of f' is negative, so f' decreases. Its sign remains to b study.

True/false. False: e^x is never zero. True if f is differentiable on the interval: this is a characterisation of convexity. False: an approximate value alone does not prove existence; the bisection must start of an interval on which continuity and a change of sign are established.

Solutions to the applications and guidance

Application 1. g is defined on $[1/2;+\infty[$ and differentiable on $]1/2;+\infty[$, with $g'(x) = 1/\sqrt{2x-1} > 0$. h is differentiable on $\mathbb{R}$, $h'(x) = -2xe^{1-x^2}$.

Application 2. The inner derivative is 3, so $f' = 12(3x+1)^3$. The coefficient x in the expansion of $(1+3x)^4$ is 12, hence $f'(0) = 12$, incompatible with the proposed formula.

Application 3. C is f, A is f' and B is f''. The zeros of A in ± 1 correspond to the extrema of C; the zero of B at 0 separates where A decreases and where it increases. Calculation: $f' = 3x^2 - 3$, $f'' = 6x$.

Application 4. All three are continuous. Polynomials are differentiable; $|x|$ has one-sided gradients -1 and 1. For x^4, $f'' = 12x^2 \geq 0$, so no inflection at 0; for x^3, $f'' = 6x$ changes sign, so inflection at 0. $|x|$ is convex on $\mathbb{R}$ and has no change of convexity at 0.

Application 5. The step function never takes the value 0: it is not continuous at 0. x^2 is continuous but not strictly monotonic on $\mathbb{R}$; $x^2 = 1$ has both solutions ± 1.

Application 6. Let $G(x) = e^{-x} - x$, continue, with $G(0) = 1 > 0$ and $G(1) = e^{-1} - 1 < 0$. $G' = -e^{-x} - 1 < 0$ ensures uniqueness. $G(1/2) > 0$, then $G(3/4) < 0$: interval $[1/2;3/4]$. From $[0;1]$, seven steps give a width of 1/128; six are not enough.

Scaffolding and review

Hide the formulas during the first work on the three curves. If the pupil confuses sign and variation, have students point to a portion of f negative but increasing. After a week, ask a new justification of uniqueness without the instruction "use the IVT".

Teaching notes and solutions

Guidance for the teacher

The progression packs this chapter into three sessions: genuinely split S29 between differentiation and convexity, then revisit the proof about tangents in S33. The vocabulary continuous/differentiable/convex must be distinguished using examples. Continuity gives existence by the IVT; the strict monotonicity gives uniqueness. For bisection, require an initial interval where the sign changes and a continuous function. A change of convexity, and not merely the vanishing of f'', characterises an inflection point.

Solutions to the activity

1. $f'(x) = 3x^2 - 3 = 3(x-1)(x+1)$ and $f''(x) = 6x$. The sign of f' is positive on $]-\infty, -1[$ and $]1, +\infty[$, zero at -1, 1, negative between these values. The sign of f'' is negative to the left of 0, zero at 0, positive to line.

2. The variations and limits are summarised by

x	$-\infty$		-1		1		$+\infty$
$f'(x)$		$+$	0	$-$	0	$+$	
$f(x)$	$-\infty$	$\nearrow$	2	$\searrow$	-2	$\nearrow$	$+\infty$

   The sign of f'' gives concavity on $]-\infty, 0]$ and convexity on $[0, +\infty[$.

3. Convexity changes at 0, and $f(0) = 0$: the point of inflection is $(0, 0)$. Its tangent has equation $y = f(0) + f'(0)x = -3x$.

4. The function is continuous. On $]-\infty, -1]$, it is strictly increasing from $-\infty$ to 2: level 1 is attained exactly once. On $[-1, 1]$, it strictly decreases from 2 to -2 : a second solution. On $[1, +\infty[$, it strictly increases from -2 à $+\infty$: a third one. The points -1, 1 are not not solutions, so no solution is counted twice.

5. Continuity guarantees a preimage for intermediate values. It does not give not uniqueness: the function studied is continuous and the level 1 has three preimages.

E1. For $u = 2x - 1$, $(u^5)' = 5u'u^4 = 10(2x-1)^4$. For $v = -x^2$, $(e^v)' = v'e^v = -2xe^{-x^2}$. Both functions are differentiable on $\mathbb{R}$.

E2. $f' = 4x^3 - 12x$ then $f'' = 12(x^2 - 1)$. The sign of f'' is positive outside of $[-1, 1]$, negative inside. Hence f is convex on $]-\infty, -1]$ and $[1, +\infty[$, concave on $[-1, 1]$. The changes of convexity give the inflection points $(-1, -5)$ and $(1, -5)$.

E3. The tangent is $y = f(1) + f'(1)(x-1) = e + e(x-1) = ex$. Since $f''(x) = e^x > 0$, the exponential is convex and its curve is above this tangent: $e^x \geq ex$ on $\mathbb{R}$.

E4. Let $g(x) = x^3 + 2x - 5$. This polynomial is continuous and $g'(x) = 3x^2 + 2 > 0$ on $\mathbb{R}$. In addition $g(1) = -2 < 0$ and $g(2) = 7 > 0$: the intermediate value theorem gives a zero in $]1, 2[$, and strictly increasing on $\mathbb{R}$ ensures that it is unique.

E5. After n steps, the length is $\frac{2}{2^n}$. The requested condition is equivalent to $2^n \geq 2000$. Since $2^{10} = 1024 < 2000 \leq 2048 = 2^{11}$, eleven steps suffice and ten do not.

Solutions to homework DM4

1. $f(0) = 1$ and $\lim_{x\to 0^+} f(x) = 1$. In $+\infty$, $f(x) = xe^{-x} + e^{-x} \to 0$ by comparative growth. By product, $f'(x) = e^{-x} - (x+1)e^{-x} = -xe^{-x}$. This derivative is strictly negative for $x > 0$: f strictly decreases from 1 to 0 on $[0, +\infty[$.

2. $f''(x) = -e^{-x} + xe^{-x} = (x-1)e^{-x}$. Hence f is concave on $[0,1]$, convex on $[1, +\infty[$. The the change of convexity at 1 gives the inflection point $(1, \frac{2}{e})$.

3. f is continuous and strictly decreasing on $[0, +\infty[$; $f(0) = 1 > \frac{1}{2}$ and its limit 0 is less to $\frac{1}{2}$. The corollary of the intermediate value theorem gives a unique solution α. In addition, $f(1) = \frac{2}{e} \simeq 0{,}7358 > \frac{1}{2}$ and $f(2) = \frac{3}{e^2} \simeq 0{,}4060 < \frac{1}{2}$, therefore $1 < \alpha < 2$.

4.
```
from maths import exp

def racine_dm4():
    a, b = 0.0, 2.0
    while b-a > 1e-3:
        m = (a+b)/2
        if (m+1)*exp(-m) > 0.5:
            a = m
        else :
            b = m
    return a, b
```

   The decrease preserves $f(a) \geq \frac{1}{2} \geq f(b)$; α remains in $[a,b]$. After eleven steps, we get $[1{,}677734375, 1{,}6787109375]$, length $\frac{1}{1024} < 10^{-3}$. The value $\alpha \simeq 1{,}67835$ is compatible with this bracketing.

5. The tangent at 0 is $y = f(0) + f'(0)x = 1$. The variations give $f(x) \leq 1$, so the whole curve is below this tangent. This does not characterise concavity: it would have to be below of all the tangents, while f becomes convex after 1.

Solutions to the textbook or past-paper work

For $f(x) = 5x^2e^x$, differentiating a product gives

$$f'(x) = 10xe^x + 5x^2e^x = 5x(x+2)e^x.$$

Since $e^x > 0$, f' is positive on $]-\infty;-2[$ and $]0;+\infty[$, negative on $]-2;0[$, zero at -2 and 0. function therefore increases up to $f(-2) = 20e^{-2}$, decreases down to $f(0) = 0$, then increases. Its limits are 0 at $-\infty$ (comparative growth) and $+\infty$ in $+\infty$. Differentiating again gives

$$f''(x) = 5(x^2+4x+2)e^x = 5\big((x+2)^2-2\big)e^x.$$

Let $a = -2-\sqrt{2}$ and $b = -2+\sqrt{2}$. The polynomial factor is positive outside of $[a;b]$, negative in $]a;b[$. Thus f is convex on $]-\infty;a]$ and $[b;+\infty[$, concave on $[a;b]$. The sign changes at both roots: the inflection points are $(a;5a^2e^a)$ and $(b;5b^2e^b)$.

The variation table can be summarised as follows:

x	$-\infty$		-2		0		$+\infty$
$f'(x)$		$+$	0	$-$	0	$+$	
$f(x)$	0	↗	$20e^{-2}$	↘	0	↗	$+\infty$

For question 5, the window below shows both changes of convexity and both extrema. The marked points correspond to the computed inflection points; the plot is used to check the consistency of the study.

